

Jusjeev Singh Bhurjee

U.S. Permanent Resident | 217-954-3761 | jusjeev1@gmail.com | linkedin.com/in/jusjeevsingh/ | github.com/Jusjeev

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

Bachelor of Science in Computer Science

08/2021 – 05/2025

- **Coursework:** Machine Learning, Computer Vision, System Programming, IoT, Numerical Methods, Computational Photography, Data Structures, Algorithms.

University of California, Davis

Davis, CA

Masters of Science in Computer Science

08/2026 – 05/2028

- Seeking internships, co-op, and early-career roles in software engineering and AI/ML.

EXPERIENCE

Agentic AI/ML Intern

01/2026 – Present

RocketFrog.ai (Agentic AI Startup)

San Ramon, CA

- Building a scalable full-stack platform combining agentic orchestration (LangGraph-based deep agents and RAG), relational databases (Postgres), knowledge graphs (Neo4j), and backend systems (Kafka, FastAPI, ayncio) to automate complex workflows.
- Collaborating with a principal scientist and CTO to refine system design and guide development based on industry use cases for businesses, developers, and research workflows.

Generative AI/ML Intern

06/2025 – 08/2025

Persistent Systems

Santa Clara, CA

- Implemented supervised LLM fine-tuning (Lora, QLoRa) on open-source SLMs for downstream tasks such as answering developer-style questions and file-path predictions, using PyTorch, Unsloth AI and Hugging Face Transformers library.
- Created fine-tuning/evaluation scripts and datasets, conducted experiments for different hyper-parameters and model combinations, and achieved 90%+ accuracy with relevant LLM benchmarks and metrics (ROUGE, F1).

Course Assistant, Computer Architecture

08/2023 – 05/2025

University of Illinois Urbana-Champaign

Champaign, IL

- Led weekly discussion sections for 250+ students covering topics including digital logic design, processor implementation, caching, parallelism, and low-level assembly programming.
- Held 1:1 and group office hours, supporting 20+ students weekly with longer programming lab assignments.

PROJECTS

Geospatial Analysis — Individual Study

08/2024 – 12/2024

- Processed multispectral satellite data (Sentinel, Copernicus, USDA) to identify crop field boundaries, and explored deep learning methods such as keypoint detection and the Segment Anything Model.
- Used the Google Earth Engine, geemap (Python), Jupyter Notebooks for data processing, experimentation, and analysis.

Farm Robotics Challenge 2025 — UIUC Team

01/2025 – 05/2025

- Built a navigation system using model predictive control and row detection for autonomous navigation of a 4-wheeled robot in straight and curved crop rows.
- Trained YOLO classification models with 90%+ accuracy for horseradish and weeds, using data collected at various angles and growth stages.
- Contributed to system development using ROS, OpenCV, PyTorch, and Roboflow, and assisted with hardware setup including Oak-d stereo camera integration and Nvidia Jetson Nano deployment.

Smart Closet System – Final Project (CS 437: IoT Systems)

01/2025 – 05/2025

- Designed and developed an IoT-enabled smart closet system with real-time wardrobe tracking and outfit recommendations.
- Used Django to integrate frontend (HTML/CSS, JavaScript) with backend (PostgreSQL) and implemented hardware components using Raspberry Pi, sensors, and ADC modules.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, TypeScript, OCaml, Verilog, MIPS Assembly

Frameworks and Libraries: FastAPI, React.js, REST APIs, SQLAlchemy, Pydantic, TanStack Query, Scikit-learn, Panda

Machine Learning/AI: PyTorch, Hugging Face Transformers, LoRA/QLoRA, YOLO, OpenCV, MediaPipe

Tools & Platforms: Git, GitHub, Linux, Jupyter, Google Earth Engine, ROS, VS Code, Claude Code

Computer Science Concepts: Object-oriented and functional programming, dynamic programming, TCP/UDP networking, supervised and deep learning, databases.